

Course 5093B

5 Credits: 4

Program Elective

Source-grounded

Speciality Polymers

□ To impart knowledge on - high performance and advanced technological applications

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5093B is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5093B.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Polymer Technology
Course code	5093B
Course title	Speciality Polymers
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4,T:0,P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart knowledge on - high performance and advanced technological applications
- of polymers.

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding of high temperature and fire-resistant polymers. Describe the structure, properties and	See official syllabus
CO2	application of polymers used in electrical and 15 Understanding electronics fields. Explain the properties and application of	See official syllabus
CO3	polymers used in construction, biomedical, 14 Understanding aerospace and telecommunication Describe the properties and application of ionomers, hydrogel, nano materials and nano 14 Understanding	See official syllabus
CO4	composites.	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	and fire-resistant polymers	4	As specified
Module 2	electrical and electronics fields.	4	As specified
Module 3	telecommunication	4	As specified
Module 4	and nano materials.	4	As specified

MODULE 1

and fire-resistant polymers

This module is presented from the official syllabus scope for Speciality Polymers. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and fire-resistant polymers
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

2 Remembering stability and fire resistance of polymers. Explain methods for improving the thermal

2 Remembering stability and fire resistance of polymers. Explain methods for improving the thermal is an official topic in Module 1 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Remembering stability and fire resistance of polymers. Explain methods for improving the thermal using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 remembering stability and fire resistance of polymers. explain methods for improving the thermal should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Remembering stability and fire resistance of polymers. Explain methods for improving the thermal means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-2 Remembering stability and fire resistance of polymers. Explain methods for improving the thermal stability of polymers. definition/meaning of thermal stability. factors affecting thermal stability, steps, application of thermal stability. Technical terms English-Malayalam.

2 Remembering stability and flame resistance of polymers. Describe the properties and high temperature

2 Remembering stability and flame resistance of polymers. Describe the properties and high temperature is an official topic in Module 1 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Remembering stability and flame resistance of polymers. Describe the properties and high temperature using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 remembering stability and flame resistance of polymers. describe the properties and high temperature should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Remembering stability and flame resistance of polymers. Describe the properties and high temperature means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Remembering stability and flame resistance of polymers. Describe the properties and high temperature stability flame resistance definition/meaning stability . stability flame resistance steps , application stability flame resistance steps . Technical terms English- stability flame resistance .

applications of polymers-PTFE, Aromatic 6 Understanding Polymers -Poly phenylene Sulphide, PPO. Explain the properties and applications of

applications of polymers-PTFE, Aromatic 6 Understanding Polymers -Poly phenylene Sulphide, PPO. Explain the properties and applications of is an official topic in Module 1 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify applications of polymers-PTFE, Aromatic 6 Understanding Polymers -Poly phenylene Sulphide, PPO. Explain the properties and applications of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, applications of polymers-ptfe, aromatic 6 understanding polymers -poly phenylene sulphide, ppo. explain the properties and applications of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What applications of polymers-PTFE, Aromatic 6 Understanding Polymers -Poly phenylene Sulphide, PPO. Explain the properties and applications of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പolymer applications of polymers-PTFE, Aromatic 6 Understanding Polymers -Poly phenylene Sulphide, PPO. Explain the properties and applications of പൊളിമറുകൾ definition/meaning പൊളിമറുകൾ. പൊളിമറുകൾ പൊളിമറുകൾ, steps, application പൊളിമറുകൾ പൊളിമറുകൾ പൊളിമറുകൾ. Technical terms English-പൊളിമറുകൾ പൊളിമറുകൾ.

PEEK, Aramid fibres, Heterocyclic 5 Understanding polymers-PBI High temperature and fire-resistant polymers improving low performance polymers for high temperature use - polymers for low fire hazards - polymers for high temperature resistance. Factors affecting the thermal stability and fire resistance of polymers - LOI- Definition Properties and applications of PTFE, Aromatic Polymers like Poly phenylene Sulphide, PPO, PEEK, Aramid, Heterocyclic polymers - polybenzimidazole (PBI) Describe the structure, properties and application of polymers used in

PEEK, Aramid fibres, Heterocyclic 5 Understanding polymers-PBI High temperature and fire-resistant polymers improving low performance polymers for high temperature use - polymers for low fire hazards - polymers for high temperature resistance. Factors affecting the thermal stability and fire resistance of polymers - LOI- Definition Properties and applications of PTFE, Aromatic Polymers like Poly phenylene Sulphide, PPO, PEEK, Aramid, Heterocyclic polymers - polybenzimidazole (PBI) Describe the structure, properties and application of polymers used in is an official topic in Module 1 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify PEEK, Aramid fibres, Heterocyclic 5 Understanding polymers-PBI High temperature and fire-resistant polymers improving low performance polymers for high temperature use - polymers for low fire hazards - polymers for high temperature resistance. Factors affecting the thermal stability and fire resistance of polymers - LOI- Definition Properties and applications of PTFE, Aromatic Polymers like Poly phenylene Sulphide, PPO, PEEK, Aramid, Heterocyclic polymers - polybenzimidazole (PBI) Describe the structure, properties and application of polymers used in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, peek, aramid fibres, heterocyclic 5 understanding polymers-pbi high temperature and fire-resistant polymers improving low performance polymers for high temperature use - polymers for low fire hazards - polymers for high temperature resistance. factors affecting the thermal stability and fire resistance of polymers – loi- definition properties and applications of ptfe, aromatic polymers like poly phenylene sulphide, ppo, peek, aramid, heterocyclic polymers - polybenzimidazole (pbi) describe the structure, properties and application of polymers used in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What PEEK, Aramid fibres, Heterocyclic 5 Understanding polymers-PBI High temperature and fire-resistant polymers improving low performance polymers for high temperature use - polymers for low fire hazards - polymers for high temperature resistance. Factors affecting the thermal stability and fire resistance of polymers – LOI- Definition Properties and applications of PTFE, Aromatic Polymers like Poly phenylene Sulphide, PPO, PEEK, Aramid, Heterocyclic polymers - polybenzimidazole (PBI) Describe the structure, properties and application of polymers used in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- PEEK, Aramid fibres, Heterocyclic 5 Understanding polymers-PBI High temperature and fire-resistant polymers improving low performance polymers for high temperature use - polymers for low fire hazards - polymers for high temperature resistance. Factors affecting the thermal stability and fire resistance of polymers – LOI- Definition Properties and applications of PTFE, Aromatic Polymers like Poly phenylene Sulphide, PPO, PEEK, Aramid, Heterocyclic polymers - polybenzimidazole (PBI) Describe the structure, properties and application of polymers used in definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

and fire-resistant polymers

	Generalize the factors affecting the thermal	
M1.01		2
Remembering		
	stability and fire resistance of polymers.	
	Explain methods for improving the thermal	
M1.02		2
Remembering		
	stability and flame resistance of polymers.	
	Describe the properties and high temperature	
M1.03	applications of polymers-PTFE, Aromatic	6
Understanding		
	Polymers -Poly phenylene Sulphide, PPO.	
	Explain the properties and applications of	
M1.04	PEEK, Aramid fibres, Heterocyclic	5
Understanding		
	polymers-PBI	

Contents:

High temperature and fire-resistant polymers improving low performance polymers for high temperature use - polymers for low fire hazards - polymers for high temperature resistance.

Factors affecting the thermal stability and fire resistance of polymers – LOI-Definition

Properties and applications of PTFE, Aromatic Polymers like Poly phenylene

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

electrical and electronics fields.

This module is presented from the official syllabus scope for Speciality Polymers. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: electrical and electronics fields.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

2 Remembering conducting mechanism. Describe the structure, properties and application of conducting polymers like

2 Remembering conducting mechanism. Describe the structure, properties and application of conducting polymers like is an official topic in Module 2 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Remembering conducting mechanism. Describe the structure, properties and application of conducting polymers like using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 remembering conducting mechanism. describe the structure, properties and application of conducting polymers like should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Remembering conducting mechanism. Describe the structure, properties and application of conducting polymers like means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഓ 2 Remembering conducting mechanism.

Describe the structure, properties and application of conducting polymers like

പോളിഅനിലിൻ പോലുള്ളവയുടെ definition/meaning എഴുതുക. പോളിഅനിലിൻ പോലുള്ളവയുടെ

steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക

എഴുതുക.

5 Understanding Polyacetylene, poly aniline, poly paraphenylene Describe the properties and application of

5 Understanding Polyacetylene, poly aniline, poly paraphenylene Describe the properties and application of is an official topic in Module 2 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding Polyacetylene, poly aniline, poly paraphenylene Describe the properties and application of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding polyacetylene, poly aniline, poly paraphenylene describe the properties and application of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding Polyacetylene, poly aniline, poly paraphenylene Describe the properties and application of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-5 Understanding Polyacetylene, poly aniline, poly paraphenylene Describe the properties and application of C_6H_4 C_6H_3 definition/meaning C_6H_4 C_6H_3 . C_6H_4 C_6H_3 C_6H_4 C_6H_3 , steps, application C_6H_4 C_6H_3 C_6H_4 C_6H_3 . Technical terms English- C_6H_4 C_6H_3 C_6H_4 C_6H_3 .

photo conducting and piezo electric polymers - 4 Understanding Poly vinyl carbazole , PVDF Explain the properties and application of Liquid crystalline polymers and Photo

photo conducting and piezo electric polymers - 4 Understanding Poly vinyl carbazole , PVDF Explain the properties and application of Liquid crystalline polymers and Photo is an official topic in Module 2 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify photo conducting and piezo electric polymers - 4 Understanding Poly vinyl carbazole , PVDF Explain the properties and application of Liquid crystalline polymers and Photo using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, photo conducting and piezo electric polymers - 4 understanding poly vinyl carbazole , pvdf explain the properties and application of liquid crystalline polymers and photo should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What photo conducting and piezo electric polymers - 4 Understanding Poly vinyl carbazole , PVDF Explain the properties and application of Liquid crystalline polymers and Photo means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പ്രകാശ സഞ്ചാലകതയും പൈയോ ഇലക്ട്രിക് പോളിമറുകൾ - 4 Understanding Poly vinyl carbazole , PVDF Explain the properties and application of Liquid crystalline polymers and Photo definition/meaning . steps, application . Technical terms English-പ്രകാശ സഞ്ചാലകതയും പൈയോ ഇലക്ട്രിക് പോളിമറുകൾ .

4 Understanding resists polymers for semiconductor application. Polymers with electrical and electronic properties Conducting polymers -definition conducting mechanisms Properties and application of Polyacetylene, Polyparaphenylene, and poly aniline. Photo conducting polymers - structure, properties application of Poly vinyl carbazole Piezo electric polymers properties - structure, properties and application PVDF Photo resist polymers for semi conductor fabrication - Liquid crystalline polymers. Explain the polymers used in construction, biomedical, aerospace and

4 Understanding resists polymers for semiconductor application. Polymers with electrical and electronic properties Conducting polymers -definition conducting mechanisms Properties and application of Polyacetylene, Polyparaphenylene, and poly aniline. Photo conducting polymers - structure, properties application of Poly vinyl carbazole Piezo electric polymers properties - structure, properties and application PVDF Photo resist polymers for semi conductor fabrication - Liquid crystalline polymers. Explain the polymers used in construction, biomedical, aerospace and is an official topic in Module 2 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding resists polymers for semiconductor application. Polymers with electrical and electronic properties Conducting polymers -definition conducting mechanisms Properties and application of Polyacetylene, Polyparaphenylene, and poly aniline. Photo conducting polymers - structure, properties application of Poly vinyl carbazole Piezo electric polymers properties - structure, properties and application PVDF Photo resist polymers for semi conductor fabrication - Liquid crystalline polymers. Explain the polymers used in construction, biomedical, aerospace and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding resists polymers for semiconductor application. polymers with electrical and electronic properties conducting polymers -definition conducting mechanisms properties and application of polyacetylene, polyparaphenylene, and poly aniline. photo conducting polymers - structure, properties application of poly vinyl carbazole piezo electric polymers properties - structure, properties and application pvdf photo resist polymers for semi conductor fabrication - liquid crystalline polymers. explain the polymers used in construction, biomedical, aerospace and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding resists polymers for semiconductor application. Polymers with electrical and electronic properties Conducting polymers -definition conducting mechanisms Properties and application of Polyacetylene, Polyparaphenylene, and poly aniline. Photo conducting polymers - structure, properties application of Poly vinyl carbazole Piezo electric polymers properties - structure, properties and application PVDF Photo resist polymers for semi conductor fabrication - Liquid crystalline polymers. Explain the polymers used in construction, biomedical, aerospace and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 4 Understanding resists polymers for semiconductor application. Polymers with electrical and electronic properties Conducting polymers -definition conducting mechanisms Properties and application of Polyacetylene, Polyparaphenylene, and poly aniline. Photo conducting polymers - structure, properties application of Poly vinyl carbazole Piezo electric polymers properties - structure, properties and application PVDF Photo resist polymers for semi conductor fabrication - Liquid crystalline polymers. Explain the polymers used in construction, biomedical, aerospace and definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

electrical and electronics fields.

M2.01	Define the conducting polymers with their	2
Remembering		

M2.02	conducting mechanism.	
Understanding	Describe the structure, properties and application of conducting polymers like	5

M2.03	Polyacetylene, poly aniline, poly paraphenylene	
Understanding	Describe the properties and application of photo conducting and piezo electric polymers -	4

M 2.04	Poly vinyl carbazole , PVDF	
Understanding	Explain the properties and application of Liquid crystalline polymers and Photo	4

	resists polymers for semiconductor application.	
	Series Test - I	1

Contents:

Polymers with electrical and electronic properties Conducting polymers -
definition
conducting mechanisms

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

telecommunication

This module is presented from the official syllabus scope for Speciality Polymers. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: telecommunication
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain the polymer concrete, types. 3 Understanding Describe the polymers and its properties

Explain the polymer concrete, types. 3 Understanding Describe the polymers and its properties is an official topic in Module 3 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the polymer concrete, types. 3 Understanding Describe the polymers and its properties using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the polymer concrete, types. 3 understanding describe the polymers and its properties should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the polymer concrete, types. 3 Understanding Describe the polymers and its properties means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പോളിമർ Explain the polymer concrete, types. 3 Understanding Describe the polymers and its properties പൊളിമർ പോളിമർ definition/meaning പൊളിമർ പോളിമർ. പൊളിമർ പോളിമർ പോളിമർ, steps, application പൊളിമർ പോളിമർ പോളിമർ. Technical terms English-പോളിമർ പോളിമർ പോളിമർ.

5 Understanding used for biomedical application. Explain the properties of polymers used as

5 Understanding used for biomedical application. Explain the properties of polymers used as is an official topic in Module 3 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding used for biomedical application. Explain the properties of polymers used as using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding used for biomedical application. explain the properties of polymers used as should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding used for biomedical application. Explain the properties of polymers used as means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-5 Understanding used for biomedical application. Explain the properties of polymers used as polymer definition/meaning polymer . polymer polymer polymer , steps, application polymer polymer polymer . Technical terms English- polymer polymer polymer .

3 Understanding binders for rocket propellants. Explain polymers used for Optical fibre for

3 Understanding binders for rocket propellants. Explain polymers used for Optical fibre for is an official topic in Module 3 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding binders for rocket propellants. Explain polymers used for Optical fibre for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding binders for rocket propellants. explain polymers used for optical fibre for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding binders for rocket propellants. Explain polymers used for Optical fibre for means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-3 Understanding binders for rocket propellants. Explain polymers used for Optical fibre for പൊളിമറുകൾ definition/meaning പൊളിമറുകൾ. പൊളിമറുകൾ പൊളിമറുകൾ, steps, application പൊളിമറുകൾ പൊളിമറുകൾ. Technical terms English-3 പൊളിമറുകൾ പൊളിമറുകൾ.

3 Understanding telecommunication cables (POFC). Polymer concrete- preparation properties, application, - PCC, PIC, PMC. Polymers for biomedical applications - PHEMA, SILICONE - properties, application Polymeric binders for rocket propellants - HTPB - properties Polymers used in Optical fibre Cable(POFC) - manufacturing, properties and application Explain the ionomers, hydrogel, Biodegradable polymers, nano polymers

3 Understanding telecommunication cables (POFC). Polymer concrete- preparation properties, application, - PCC, PIC, PMC. Polymers for biomedical applications - PHEMA, SILICONE - properties, application Polymeric binders for rocket propellants - HTPB - properties Polymers used in Optical fibre Cable(POFC) - manufacturing, properties and application Explain the ionomers, hydrogel, Biodegradable polymers, nano polymers is an official topic in Module 3 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding telecommunication cables (POFC). Polymer concrete- preparation properties, application, - PCC, PIC, PMC. Polymers for biomedical applications - PHEMA, SILICONE - properties, application Polymeric binders for rocket propellants - HTPB - properties Polymers used in Optical fibre Cable(POFC) - manufacturing, properties and application Explain the ionomers, hydrogel, Biodegradable polymers, nano polymers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding telecommunication cables (pofc). polymer concrete- preparation properties, application, - pcc, pic, pmc. polymers for biomedical applications - pHEMA, silicone - properties, application polymeric binders for rocket propellants - htpb - properties polymers used in optical fibre cable(pofc) -

manufacturing, properties and application explain the ionomers, hydrogel, biodegradable polymers, nano polymers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding telecommunication cables (POFC). Polymer concrete- preparation properties, application, - PCC, PIC, PMC. Polymers for biomedical applications - PHEMA, SILICONE - properties, application Polymeric binders for rocket propellants - HTPB - properties Polymers used in Optical fibre Cable(POFC) – manufacturing, properties and application Explain the ionomers, hydrogel, Biodegradable polymers, nano polymers means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 Understanding telecommunication cables (POFC). Polymer concrete- preparation properties, application, - PCC, PIC, PMC. Polymers for biomedical applications - PHEMA, SILICONE - properties, application Polymeric binders for rocket propellants - HTPB - properties Polymers used in Optical fibre Cable(POFC) – manufacturing, properties and application Explain the ionomers, hydrogel, Biodegradable polymers, nano polymers definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

telecommunication

M3.01 Explain the polymer concrete, types. 3
Understanding

Describe the polymers and its properties

M3.02 5
Understanding

used for biomedical application.

Explain the properties of polymers used as

M3.03 3
Understanding

binders for rocket propellants.

M3.04 3
Understanding

Explain polymers used for Optical fibre for

telecommunication cables (POFC).

Contents:

Polymer concrete- preparation properties, application, - PCC, PIC, PMC.

Polymers for biomedical applications - PHEMA, SILICONE - properties, application

Polymeric binders for rocket propellants - HTPB - properties

Polymers used in Optical fibre Cable(POFC) – manufacturing, properties and application

Explain the ionomers, hydrogel, Biodegradable polymers, nano polymers

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

and nano materials.

This module is presented from the official syllabus scope for Speciality Polymers. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and nano materials.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

2 Remembering electrolytes and hydrogel. Describe the Synthesis, physical

2 Remembering electrolytes and hydrogel. Describe the Synthesis, physical is an official topic in Module 4 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Remembering electrolytes and hydrogel. Describe the Synthesis, physical using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 remembering electrolytes and hydrogel. describe the synthesis, physical should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Remembering electrolytes and hydrogel. Describe the Synthesis, physical means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4- 2 Remembering electrolytes and hydrogel.

Describe the Synthesis, physical properties definition/meaning
properties. properties properties, steps, application properties properties
properties. Technical terms English- properties properties.

properties and applications of ionomers 4 Understanding based on PS, PTFE

Explain the properties and applications of

properties and applications of ionomers 4 Understanding based on PS, PTFE Explain the properties and applications of is an official topic in Module 4 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify properties and applications of ionomers 4 Understanding based on PS, PTFE Explain the properties and applications of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, properties and applications of ionomers 4 understanding based on ps, ptfe explain the properties and applications of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What properties and applications of ionomers 4 Understanding based on PS, PTFE Explain the properties and applications of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഈ പ്രോപ്പർട്ടീസുകളും അപ്ലിക്കേഷനുകളും ഓഫ് ഓണമേഴ്സ് 4
Understanding based on PS, PTFE Explain the properties and applications of
ഓണമേഴ്സ് ഓണമേഴ്സ് definition/meaning ഓണമേഴ്സ്. ഓണമേഴ്സ് ഓണമേഴ്സ് ഓണമേഴ്സ്,
steps, application ഓണമേഴ്സ് ഓണമേഴ്സ് ഓണമേഴ്സ്. Technical terms English-ഈ ഓണമേഴ്സ്
ഓണമേഴ്സ്.

4 Understanding biodegradable polymers, PLA, Chitosan. Generalize the idea of nano polymers,

4 Understanding biodegradable polymers, PLA, Chitosan. Generalize the idea of nano polymers, is an official topic in Module 4 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding biodegradable polymers, PLA, Chitosan. Generalize the idea of nano polymers, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding biodegradable polymers, pla, chitosan. generalize the idea of nano polymers, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding biodegradable polymers, PLA, Chitosan. Generalize the idea of nano polymers, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-4 Understanding biodegradable polymers, PLA, Chitosan. Generalize the idea of nano polymers, നാനോ പോളിമറുകൾ definition/meaning നാനോ പോളിമറുകൾ. നാനോ പോളിമറുകൾ, steps, application നാനോ പോളിമറുകൾ. Technical terms English-നാനോ പോളിമറുകൾ.

preparation, properties and application of 4 Understanding nano composites
Properties of ionic polymers Ion - exchange, Hydrophilicity. Poly electrolytes-
definition Synthesis, physical properties and applications of ionomers based
on PS, PTFE, Biodegradable polymers - Properties and applications of PLA,
chitosan. Polymer Nano composite - Definition, properties and application of
nano composites Text / Reference: T/R Book Title/Author H.F.Mark, (Ed),
Encyclopedia of polymer Science & Engineering, John Wiley & T1 Sons, New
York, 1989. R1 Matrin.T.Goosey, Plastics for Electronics, Elsevier, Applied
Science, 1985. Manas Chanda, Salil.K.Roy, Plastics Technology Hand book,
2nd edition, Marcel R2 Dekker, New York, 1993 R3 R.W. Dyson, Specialty
Polymers, Chapman & Hall, 2nd edition, 1998. Joseph H Koo, Fundamentals
properties and applications of polymer nano R4 composites, Cambridge
University Press, 2016. Online Resources: Sl.No Website Link 1
<https://www.researchgate.net/> 2 <https://onlinelibrary.wiley.com> 3
<https://www.ncbi.nlm.nih.gov>

preparation, properties and application of 4 Understanding nano composites
Properties of ionic polymers Ion - exchange, Hydrophilicity. Poly electrolytes-
definition Synthesis, physical properties and applications of ionomers based on
PS, PTFE, Biodegradable polymers - Properties and applications of PLA, chitosan.
Polymer Nano composite - Definition, properties and application of nano
composites Text / Reference: T/R Book Title/Author H.F.Mark, (Ed), Encyclopedia of
polymer Science & Engineering, John Wiley & T1 Sons, New York, 1989. R1
Matrin.T.Goosey, Plastics for Electronics, Elsevier, Applied Science, 1985. Manas
Chanda, Salil.K.Roy, Plastics Technology Hand book, 2nd edition, Marcel R2
Dekker, New York, 1993 R3 R.W. Dyson, Specialty Polymers, Chapman & Hall, 2nd
edition, 1998. Joseph H Koo, Fundamentals properties and applications of
polymer nano R4 composites, Cambridge University Press, 2016. Online
Resources: Sl.No Website Link 1 <https://www.researchgate.net/> 2
<https://onlinelibrary.wiley.com> 3 <https://www.ncbi.nlm.nih.gov> is an official topic in
Module 4 of Speciality Polymers. Study the terminology first, then connect the
stated purpose, sequence, components or application to the wider course
outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify preparation, properties and application of 4 Understanding nano composites Properties of ionic polymers Ion - exchange, Hydrophilicity. Poly electrolytes- definition Synthesis, physical properties and applications of ionomers based on PS, PTFE, Biodegradable polymers - Properties and applications of PLA, chitosan. Polymer Nano composite - Definition, properties and application of nano composites Text / Reference: T/R Book Title/Author H.F.Mark, (Ed), Encyclopedia of polymer Science & Engineering, John Wiley & T1 Sons, New York, 1989. R1 Matrin.T.Goosey, Plastics for Electronics, Elsevier, Applied Science, 1985. Manas Chanda, Salil.K.Roy, Plastics Technology Hand book, 2nd edition, Marcel R2 Dekker, New York, 1993 R3 R.W. Dyson, Specialty Polymers, Chapman & Hall, 2nd edition, 1998. Joseph H Koo, Fundamentals properties and applications of polymer nano R4 composites, Cambridge University Press, 2016. Online Resources: Sl.No Website Link 1 <https://www.researchgate.net/> 2 <https://onlinelibrary.wiley.com> 3 <https://www.ncbi.nlm.nih.gov> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, preparation, properties and application of 4 understanding nano composites properties of ionic polymers ion - exchange, hydrophilicity. poly electrolytes- definition synthesis, physical properties and applications of ionomers based on ps, ptfe, biodegradable polymers - properties and applications of pla, chitosan. polymer nano composite - definition, properties and application of nano composites text / reference: t/r book title/author h.f.mark, (ed), encyclopedia of polymer science & engineering, john wiley & t1 sons, new york, 1989. r1 matrin.t.goosey, plastics for electronics, elsevier, applied science, 1985. manas chanda, salil.k.roy, plastics technology hand book, 2nd edition, marcel r2 dekker, new york, 1993 r3 r.w. dyson, specialty polymers, chapman & hall, 2nd edition, 1998. joseph h koo, fundamentals properties and applications of polymer nano r4 composites, cambridge university press, 2016. online resources: sl.no website link 1 <https://www.researchgate.net/> 2 <https://onlinelibrary.wiley.com> 3 <https://www.ncbi.nlm.nih.gov> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What preparation, properties and application of 4 Understanding nano composites Properties of ionic polymers Ion - exchange, Hydrophilicity. Poly electrolytes- definition Synthesis, physical properties and applications of ionomers based on PS, PTFE, Biodegradable polymers - Properties and applications of PLA, chitosan. Polymer Nano composite - Definition, properties and application of nano composites Text / Reference: T/R Book Title/Author H.F.Mark, (Ed), Encyclopedia of polymer Science & Engineering, John Wiley & T1 Sons, New York, 1989. R1 Matrin.T.Goosey,

Aspect	What to prepare
	Plastics for Electronics, Elsevier, Applied Science, 1985. Manas Chanda, Salil.K.Roy, Plastics Technology Hand book, 2nd edition, Marcel R2 Dekker, New York, 1993 R3 R.W. Dyson, Specialty Polymers, Chapman & Hall, 2nd edition, 1998. Joseph H Koo, Fundamentals properties and applications of polymer nano R4 composites, Cambridge University Press, 2016. Online Resources: Sl.No Website Link 1 https://www.researchgate.net/ 2 https://onlinelibrary.wiley.com 3 https://www.ncbi.nlm.nih.gov means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ preparation, properties and application of 4 Understanding nano composites Properties of ionic polymers Ion - exchange, Hydrophilicity. Poly electrolytes- definition Synthesis, physical properties and applications of ionomers based on PS, PTFE, Biodegradable polymers - Properties and applications of PLA, chitosan. Polymer Nano composite - Definition, properties and application of nano composites Text / Reference: T/R Book Title/Author H.F.Mark, (Ed), Encyclopedia of polymer Science & Engineering, John Wiley & T1 Sons, New York, 1989. R1 Matrin.T.Goosey, Plastics for Electronics, Elsevier, Applied Science, 1985. Manas Chanda, Salil.K.Roy, Plastics Technology Hand book, 2nd edition, Marcel R2 Dekker, New York, 1993 R3 R.W. Dyson, Specialty Polymers, Chapman & Hall, 2nd edition, 1998. Joseph H Koo, Fundamentals properties and applications of polymer nano R4 composites, Cambridge University Press, 2016. Online Resources: Sl.No Website Link 1 <https://www.researchgate.net/> 2 <https://onlinelibrary.wiley.com> 3 <https://www.ncbi.nlm.nih.gov> □□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□ □□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

and nano materials.

M4.01	Define the ionic polymers, poly electrolytes and hydrogel.	2
Remembering		
M4.02	Describe the Synthesis, physical properties and applications of ionomers based on PS, PTFE	4
Understanding		
M4.03	Explain the properties and applications of biodegradable polymers, PLA, Chitosan.	4
Understanding		
M4.04	Generalize the idea of nano polymers, preparation, properties and application of nano composites	4
Understanding		
	Series Test - II	1

Contents:

Properties of ionic polymers Ion - exchange, Hydrophilicity. Poly electrolytes-definition

Synthesis, physical properties and applications of ionomers based on PS, PTFE, Biodegradable polymers - Properties and applications of PLA, chitosan.

Polymer Nano composite - Definition, properties and application of nano composites

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 2 Remembering stability and fire resistance of polymers. Explain methods for improving the thermal. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Remembering stability and fire resistance of polymers*. Explain methods for improving the thermal. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 2 Remembering stability and flame resistance of polymers. Describe the properties and high temperature. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Remembering stability and flame resistance of polymers. Describe the properties and high temperature.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain applications of polymers-PTFE, Aromatic 6 Understanding Polymers -Poly phenylene Sulphide, PPO. Explain the properties and applications of. (3 marks)

Answer: Begin with the standard definition or identity of *applications of polymers-PTFE, Aromatic 6 Understanding Polymers -Poly phenylene Sulphide, PPO. Explain the properties and applications of.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain PEEK, Aramid fibres, Heterocyclic 5 Understanding polymers-PBI High temperature and fire-resistant polymers improving low performance polymers for high temperature use - polymers for low fire hazards - polymers for high temperature resistance. Factors affecting the thermal stability and fire resistance of polymers - LOI- Definition Properties and applications of PTFE, Aromatic Polymers like Poly phenylene Sulphide, PPO, PEEK, Aramid, Heterocyclic polymers - polybenzimidazole (PBI) Describe the structure, properties and application of polymers used in. (3 marks)

Answer: Begin with the standard definition or identity of *PEEK, Aramid fibres, Heterocyclic 5 Understanding polymers-PBI High temperature and fire-resistant polymers improving low performance polymers for high temperature use - polymers for low fire hazards - polymers for high temperature resistance. Factors affecting the thermal stability and fire resistance of polymers - LOI- Definition Properties and applications of PTFE, Aromatic Polymers like Poly phenylene Sulphide, PPO, PEEK, Aramid, Heterocyclic polymers - polybenzimidazole (PBI) Describe the structure, properties and application of polymers used in.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain 2 Remembering conducting mechanism. Describe the structure, properties and application of conducting polymers like. (3 marks)

Answer: Begin with the standard definition or identity of *2 Remembering conducting mechanism. Describe the structure, properties and application of conducting polymers like.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 5 Understanding Polyacetylene, poly aniline, poly paraphenylene Describe the properties and application of. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding Polyacetylene, poly aniline, poly paraphenylene Describe the properties and application of.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain photo conducting and piezo electric polymers - 4 Understanding Poly vinyl carbazole , PVDF Explain the properties and application of Liquid crystalline polymers and Photo. (3 marks)

Answer: Begin with the standard definition or identity of *photo conducting and piezo electric polymers - 4 Understanding Poly vinyl carbazole , PVDF Explain the properties and application of Liquid crystalline polymers and Photo*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding resists polymers for semiconductor application. Polymers with electrical and electronic properties Conducting polymers -definition conducting mechanisms Properties and application of Polyacetylene, Polyparaphenylene, and poly aniline. Photo conducting polymers - structure, properties application of Poly vinyl carbazole Piezo electric polymers properties - structure, properties and application PVDF Photo resist polymers for semi conductor fabrication - Liquid crystalline polymers. Explain the polymers used in construction, biomedical, aerospace and. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding resists polymers for semiconductor application. Polymers with electrical and electronic properties Conducting polymers -definition conducting mechanisms Properties and application of Polyacetylene, Polyparaphenylene, and poly aniline. Photo conducting polymers - structure, properties application of Poly vinyl carbazole Piezo electric polymers properties - structure, properties and application PVDF Photo resist polymers for semi conductor fabrication - Liquid crystalline polymers. Explain the polymers used in construction, biomedical, aerospace and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps,

□□□□□ application □□□□ □□□□□□□□ □□□□□.

Module 3

Define or explain Explain the polymer concrete, types. 3 Understanding
Describe the polymers and its properties. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the polymer concrete, types. 3 Understanding Describe the polymers and its properties.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 5 Understanding used for biomedical application. Explain the properties of polymers used as. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding used for biomedical application*. Explain the properties of polymers used as. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding binders for rocket propellants. Explain polymers used for Optical fibre for. (3 marks)

Answer: Begin with the standard definition or identity of 3 *Understanding binders for rocket propellants*. Explain *polymers used for Optical fibre* for. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 3 Understanding telecommunication cables (POFC). Polymer concrete- preparation properties, application, - PCC, PIC, PMC. Polymers for biomedical applications - PHEMA, SILICONE - properties, application Polymeric binders for rocket propellants - HTPB - properties Polymers used in Optical fibre Cable(POFC) - manufacturing, properties and application Explain the ionomers, hydrogel, Biodegradable polymers, nano polymers. (3 marks)

Answer: Begin with the standard definition or identity of 3 *Understanding telecommunication cables (POFC). Polymer concrete- preparation properties, application, - PCC, PIC, PMC. Polymers for biomedical applications - PHEMA, SILICONE - properties, application Polymeric binders for rocket propellants - HTPB - properties Polymers used in Optical fibre Cable(POFC) - manufacturing, properties and application Explain the ionomers, hydrogel, Biodegradable polymers, nano polymers.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 4

Define or explain 2 Remembering electrolytes and hydrogel. Describe the Synthesis, physical. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Remembering electrolytes and hydrogel. Describe the Synthesis, physical.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain properties and applications of ionomers 4 Understanding based on PS, PTFE Explain the properties and applications of. (3 marks)

Answer: Begin with the standard definition or identity of *properties and applications of ionomers 4 Understanding based on PS, PTFE Explain the properties and applications of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding biodegradable polymers, PLA, Chitosan. Generalize the idea of nano polymers,. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding biodegradable polymers, PLA, Chitosan. Generalize the idea of nano polymers,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain preparation, properties and application of 4 Understanding nano composites Properties of ionic polymers Ion - exchange, Hydrophilicity. Poly electrolytes- definition Synthesis, physical properties and applications of ionomers based on PS, PTFE, Biodegradable polymers - Properties and applications of PLA, chitosan. Polymer Nano composite - Definition, properties and application of nano composites Text / Reference: T/R Book Title/Author H.F.Mark, (Ed), Encyclopedia of polymer Science & Engineering, John Wiley & T1 Sons, New York, 1989. R1 Matrin.T.Goosey, Plastics for Electronics, Elsevier, Applied Science, 1985. Manas Chanda, Salil.K.Roy, Plastics Technology Hand book, 2nd edition, Marcel R2 Dekker, New York, 1993 R3 R.W. Dyson, Specialty Polymers, Chapman & Hall, 2nd edition, 1998. Joseph H Koo, Fundamentals properties and applications of polymer nano R4 composites, Cambridge University Press, 2016. Online Resources: Sl.No Website Link 1

<https://www.researchgate.net/> 2 <https://onlinelibrary.wiley.com> 3
<https://www.ncbi.nlm.nih.gov>. (3 marks)

Answer: Begin with the standard definition or identity of *preparation, properties and application of 4 Understanding nano composites Properties of ionic polymers Ion - exchange, Hydrophilicity. Poly electrolytes- definition Synthesis, physical properties and applications of ionomers based on PS, PTFE, Biodegradable polymers - Properties and applications of PLA, chitosan. Polymer Nano composite - Definition, properties and application of nano composites Text / Reference: T/R Book Title/Author H.F.Mark, (Ed), Encyclopedia of polymer Science & Engineering, John Wiley & T1 Sons, New York, 1989. R1 Matrin.T.Goosey, Plastics for Electronics, Elsevier, Applied Science, 1985. Manas Chanda, Salil.K.Roy, Plastics Technology Hand book, 2nd edition, Marcel R2 Dekker, New York, 1993 R3 R.W. Dyson, Specialty Polymers, Chapman & Hall, 2nd edition, 1998. Joseph H Koo, Fundamentals properties and applications of polymer nano R4 composites, Cambridge University Press, 2016. Online Resources: Sl.No Website Link 1 <https://www.researchgate.net/> 2 <https://onlinelibrary.wiley.com> 3 <https://www.ncbi.nlm.nih.gov>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.*

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Remembering stability and fire resistance of polymers. Explain methods for improving the thermal.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **2 Remembering conducting mechanism. Describe the structure, properties and application of conducting polymers like.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain the polymer concrete, types. 3 Understanding Describe the polymers and its properties.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **2 Remembering electrolytes and hydrogel. Describe the Synthesis, physical.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://www.researchgate.net/> <https://onlinelibrary.wiley.com>
<https://www.ncbi.nlm.nih.gov>

IN-PAGE SEARCH

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